



JAIPUR NATIONAL UNIVERSITY

Semester/Year End Examination May-2023

Name of School: School of Engineering and Technology

Name of Programme/Course: B.Tech. (Biomedical) / B.Tech. (Chemical) / B.Tech. (Civill) / B.Tech. (Mechanical) / B.Tech. (CSE) / B.Tech. (CSE-AIML) / B.Tech. (CSE-Cyber Security) / B.Tech. (Electrical)

Semester: II

Paper Code: BTEMACO204T22

Title of Paper: Engineering Mathematics-II

Time: 3 hrs.

Max. Marks: 60

INSTRUCTIONS/NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section A and Section B. Section-A consists of seven questions and Section-B consists of six question. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question
Number

SECTION-A

(Attempt any five questions.)

Marks
Allotted

Q.1.

Find the Eigen values of $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 0 \\ 5 & 0 & 3 \end{bmatrix}$.

(4)

Q.2.

Prove the Euler formula of a_0 and a_n of Fourier series for the function $f(x)$.

(4)

Q.3.

Find the value of a_n only of Fourier series for $f(x) = x^2$ in the interval $(-\pi, \pi)$.

(4)

Q.4. (a) Solve: $\frac{dy}{dx} = e^{(x-y)} + x^2 e^{-y}$

(2*2)

(b) Solve: $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = 0$

- Q.5. Solve $x \frac{dy}{dx} = \sqrt{x^2 + 1}$
- Q.6. Solve: $(D^2 + a^2) y = \sec(ax)$.
- Q.7. Solve $pt - qs = q^3$.

SECTION-B

(Attempt any four questions.)

- Q.1 Find the rank of A by Normal form

(10)

$$A = \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$$

Q.2.

Verify Cayley Hamilton theorem of $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$.

(10)

Also find A^{-1} .

Q.3.

Find the value of a_0, a_n of Fourier series of the function

$f(x) = x + x^2$ in the interval $(-\pi, \pi)$.

(10)

Q.4.

Solve: $(D^2 - 4D + 4) y = 8x^2 e^{2x} \sin 2x$.

(10)

Q.5.

Solve: $x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + y = \frac{1}{(1-x)^2}$.

(10)

Q.6.

Find the complete integral of $p^2 x^2 + q^2 y^2 = z$.

(10)

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$f(x) = x + x^2$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} (x + x^2) dx = \frac{1}{2\pi} \left[\frac{x^2}{2} + \frac{x^3}{3} \right]_{-\pi}^{\pi} = \frac{1}{2\pi} \left(\frac{\pi^2}{2} + \frac{\pi^3}{3} - \left(\frac{\pi^2}{2} - \frac{\pi^3}{3} \right) \right) = \frac{1}{2\pi} \left(\frac{2\pi^3}{3} \right) = \frac{\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \cos nx dx = \frac{1}{\pi} \left(\int_{-\pi}^{\pi} x \cos nx dx + \int_{-\pi}^{\pi} x^2 \cos nx dx \right)$$

$$= \frac{1}{\pi} \left(\left[\frac{x \sin nx}{n} - \frac{\sin nx}{n^2} \right]_{-\pi}^{\pi} + \left[\frac{x^2 \sin nx}{n} - \frac{2x \cos nx}{n^2} + \frac{2 \sin nx}{n^3} \right]_{-\pi}^{\pi} \right) = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \sin nx dx = \frac{1}{\pi} \left(\int_{-\pi}^{\pi} x \sin nx dx + \int_{-\pi}^{\pi} x^2 \sin nx dx \right)$$

$$= \frac{1}{\pi} \left(\left[-\frac{x \cos nx}{n} - \frac{\sin nx}{n^2} \right]_{-\pi}^{\pi} + \left[\frac{x^2 \cos nx}{n} - \frac{2x \sin nx}{n^2} - \frac{2 \cos nx}{n^3} \right]_{-\pi}^{\pi} \right) = \frac{1}{\pi} \left(-\frac{\pi \cos n\pi}{n} - \frac{\sin n\pi}{n^2} + \frac{\pi \cos n\pi}{n} + \frac{\sin n\pi}{n^2} \right) = 0$$